SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	ECE	**Date:**	27.08.26

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization
   A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
2. Cloud Server Memory Bottleneck
   A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
3. Autonomous Vehicle Data Processing
   An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
4. AI Inference Edge Device
   An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
5. High-Performance Gaming Console
   A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

***Code of Conduct**:*

I GOPIKA P (192412306) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:GOPIKA P

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	**Application of Engineering Concepts (25%)**	**Solution Design (25%)**	**Use of Tools (15%)**	**Analysis (10%)**
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

ANSWER

1. Smartphone Performance Optimization.

	L1 Cache	**L2 Cache**	**L3 Cache**	**DRAM**
Latency	~1–2 cycles (fastest)	~3–10 cycles	~30–50 cycles	~100 ns
Bandwidth	Very high	High	Moderate	High (DDR/LPDDR)
Role	Immediate instruction/data reuse	Mid-level reuse, balances speed/size	Shared across cores, reduces main memory calls	Bulk working memory for apps
Problem	Lag during app switching due to DRAM bottleneck			
Redesign	Increase L2/L3 size, optimize DRAM bandwidth	Use prefetching + larger caches	Faster DRAM (LPDDR5)	Hierarchical balance reduces latency

Conclusion: Smartphone lag arises when DRAM access dominates over cache hits. By enlarging L2/L3 caches and adopting faster DRAM (LPDDR5/LPDDR5X), latency can be reduced and throughput improved. A hierarchical design where L1 handles immediate reuse, L2/L3 absorb mid-range requests, and DRAM provides high-bandwidth storage ensures smoother app switching and better user experience.

2. Cloud Server Memory Bottleneck

Aspect	Cache Hierarchy	Virtual Memory Paging	Recommended Tech
Latency	Nanoseconds	Microseconds (disk/SSD access)	NVMe SSD (NAND Flash) + DRAM
Bandwidth	High	Limited	High throughput with DRAM + SSD
Role	Speeds up query execution	Handles large datasets beyond RAM	Hybrid: larger L3 + DRAM + SSD
Redesign	Expand L3 cache, reduce paging	Use NVMe NAND Flash for swap	Cache-aware DB design

Conclusion: Database query latency stems from excessive paging to slower storage. Strengthening cache hierarchy and minimizing paging through NVMe SSDs and larger DRAM pools reduces query response times. A hybrid memory system combining cache, DRAM, and NAND Flash ensures high throughput and scalable performance for cloud workloads.

3. Autonomous Vehicle Data Processing

Aspect	**SRAM**	**DRAM**	**MRAM**
Latency	~1 ns (fastest)	~10–100 ns	~10 ns
Bandwidth	Very high	High	Moderate
Volatility	Volatile	Volatile	Non-volatile
Role	Sensor buffers, real-time compute	Bulk working memory	Persistent logs, fault tolerance
Problem	Real-time sensor data needs low latency		
Redesign	SRAM for immediate sensor data	DRAM for compute workloads	MRAM for persistent storage

Conclusion: Autonomous vehicles demand real-time responsiveness. SRAM provides ultra-low latency for sensor buffering, DRAM supports high-bandwidth computation, and MRAM ensures non-volatile persistence for safety logs. This layered hierarchy balances speed, bandwidth, and reliability, enabling safe and efficient autonomous driving.

4. AI Inference Edge Device

Aspect	**NAND Flash**	**DRAM**	**RRAM**
Latency	Microseconds	~100 ns	~10 ns
Bandwidth	Moderate	High	High
Power	Low	Moderate	Very low
Role	Long-term storage	Working memory	Store model weights, inference
Problem	Fast model loading with low power		
Redesign	NAND Flash for storage	DRAM for execution	RRAM for inference weights

Conclusion: Edge AI devices require low-power yet fast memory. NAND Flash provides cost-effective long-term storage, DRAM enables high-speed execution, and RRAM offers ultra-low-power inference weight storage. This hierarchy accelerates model loading while minimizing energy consumption, making AI inference efficient on constrained edge hardware.

5. High-Performance Gaming Console

Aspect	**L3 Cache**	**DRAM (e.g., GDDR6)**
Latency	~30–50 cycles	~100 ns
Bandwidth	Moderate	Very high (hundreds of GB/s)
Role	Shared cache for CPU cores	Frame buffer + scene data
Problem	Frame drops during scene loading	
Redesign	Larger L3 cache for CPU	High-bandwidth GDDR6 DRAM

Conclusion: Gaming consoles suffer frame drops when DRAM throughput cannot keep up with scene complexity. Enlarging L3 cache reduces CPU stalls, while adopting high-bandwidth DRAM (GDDR6/GDDR6X) ensures smooth rendering. A balanced hierarchy between cache and DRAM enhances frame stability and delivers immersive gameplay.